



Date: 06.04.2026

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CONTINUOUS INTERNAL ASSESSMENT – II

Fourth Semester (Regulation 2021)

B. Tech / Artificial Intelligence and Machine Learning

21AM421 – DEEP LEARNING

Time: 1 hour 30 minutes

Maximum: 50 Marks

Answer all the Questions

Part A - (12*0.5=6 Marks)

- _____ network is designed to reduce overfitting by penalizing the Jacobian of the encoder output.
a) Sparse Autoencoder b) Denoising Autoencoder
c) Contractive Autoencoder d) VAE
- A Deep Boltzmann Machine is a generative model composed of multiple layers of _____ variables.
- Say True or False: Standard autoencoders always learn a probabilistic latent space.
- The encoder in a sequence-to-sequence model is responsible for:
a) Predicting output directly b) Mapping output to input
c) Summarizing input sequence d) Generating noise
- Say True or False: Adversarial training involves only supervised learning principles.
- GRU architecture simplifies LSTM by merging the _____ and input gates.
- _____ technique is used for separating an image into distinct regions based on object boundaries.
a) Captioning b) Segmentation c) Detection d) Translation
- Say True or False: Recursive Neural Networks are typically used for sentiment analysis at the sentence and phrase level.
- _____ generation models use LSTM networks to produce text responses in a conversation.
- _____ task uses both CNN and LSTM together to generate textual context from visuals.
a) Video classification b) Image captioning c) Object tracking
d) Opinion mining

- Say True or False: Convolutional Neural Networks are used for Named Entity Recognition tasks.
- In opinion mining, recurrent architectures help to retain _____ across word sequences.

Answer any two questions

Part B - (3*2=6 Marks)

- Compare the performance of standard Autoencoders vs. Sparse Autoencoders on high-dimensional data.
- Determine the suitability of Generative Adversarial Networks (GANs) for real-world applications such as image enhancement and generation.
- Interpret the impact of attention mechanisms in improving the performance of image captioning models.

Answer any two questions

Part C - (2*14=28 Marks)

- Evaluate the suitability of GRU architecture for time-series forecasting by examining its structure and ability to model sequential dependencies.
- Demonstrate the implementation of Named Entity Recognition (NER) using deep learning, and explain the steps involved in identifying entities in a text.
- Apply an LSTM model to generate captions for video frames after preprocessing the video dataset.

Answer the questions

Part D - (1*10=10 Marks)

- a) Demonstrate the implementation of a Denoising Autoencoder for image reconstruction by outlining the key steps in training and testing the model with noisy image data.
(OR)
- b) Develop a deep learning model using the VGG16 pre-trained architecture to classify images of fruits into three categories: Apple, Banana, and Orange.

Staff I/C(s)

PAC

HOD

IQAC

CO1:	Understand the role of Deep learning in Machine Learning Applications.
CO2:	To get familiar with the use of TensorFlow/Keras in Deep Learning Applications.
CO3:	To design and implement Deep Learning Applications.
CO4:	Critically analyse Different Deep Learning Models in Image Related Projects.
CO5:	To design and implement Convolutional Neural Networks.
CO6:	To know about applications of Deep Learning in NLP and Image Processing

Levels: Understanding – U, Remember – R, Applying – A, Analyze - AN, E- Evaluate, C- Create

Cos / Level	CO1	CO2	CO3	CO4	CO5	CO6	Marks
R				1,2,4,6	8,9		3
U				3,5	7,10,11,12,15		5
A				19(a)	17	18	38
AN				13	14		4
E				16			14
C						19(b)	10
Marks				31	19	24	74

Staff I/C(s)

PAC

HOD

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